

AI 504 Problem set 4

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Due Friday, April 17

Instructions. These are identical to preceding problem sets. Summary: you are allowed/encouraged to submit in groups; all students responsible for all problems; no adaptive online help.

Grading. A grade of 7 or higher means that you have *at understood all of the questions*. Even in the case that you do not completely solve a problem, write down your most coherent and intelligent response. On top of which, a 10 means a correct answer to at least three questions, and a 12 means a correct answer to all questions. Grades of 0, 2, or 4 are awarded on an ad-hoc basis.

The problems in this problem set are mostly due to Larry Moss.

1. Do exercise 2.3 from the textbook, reproduced herewith. Work with the set of nouns hawks, birds, turtles and the verb see. Let Γ be the set of sentences

- All hawks birds
- All (see all turtles) (see all (see all hawks))
- All (see all (see all hawks)) turtles
- All turtles (see all birds)

Give a formal proof (in either $A(RC)$ or $A(RC)'$) of each of the following sentences from Γ . (Attach more pages if space demands.)

- (a) All (see all (see all hawks)) (see all hawks)
- (b) All (see all turtles) (see all (see all birds))
- (c) All (see all turtles) (see all (see all turtles))

2. This problem lets you practice *induction on terms*. Work with the set of verbs see, love and one noun cat. We get terms like

love all (see all (see all (love all (love all cat)))).

This term starts with one occurrence of love. The terms cat and see all (love all cat) start with zero occurrences of love. And love all (love all cat)) starts with two occurrences of love.

Let's take a model $\mathcal{M}$ with $M = \{1, 2\}$ and with

$$\llbracket \text{cat} \rrbracket = \{1, 2\}$$

$$\llbracket \text{see} \rrbracket = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$$

$$\llbracket \text{love} \rrbracket = \{(1, 2)\}$$

Prove that for any term t , the denotation $\llbracket t \rrbracket \subseteq M$ is either M itself or $\emptyset$, depending on whether the number of occurrences of love that t starts with is either even or odd.

3. Let $\Gamma = \{\text{All } p \text{ } q\}$. For each of the following two sentences give *either* a proof (in either A(RC) or A(RC)') of that sentence from Γ *or* a counter-model (i.e., a model which satisfies All $p \text{ } q$ but falsifies the sentence in question). If you're giving a counter-model, verify that your counter-model actually works, i.e., satisfies and falsifies the right sentences, and show your work.

(a) All (hate all (love all p)) (hate all (love all q))

(b) All (hate all (love all q)) (hate all (love all p)).

4. This problem continues the notion of duals from homework 2. We can define the *dual* of the sentence $\text{All } s \text{ } t$ to be the sentence $\text{All } t \text{ } s$. So for example, the dual of the sentence $\text{All turtles (see all birds)}$ is $\text{All (see all birds) turtles}$. Let φ^+ denote the dual of φ . Note that this *extends* the previous notion of dual to sentences in $\mathcal{A}(\mathcal{RC})$.

My question is: if $\Gamma \vdash \varphi$, is it still true that $\Gamma^+ \vdash \varphi^+$? You must either say **yes** and include a proof by induction on proof trees (like before), or **no** and a counterexample. (A counterexample consists of a *specific* set of sentences Γ and a sentence φ and a *specific* model $\mathcal{M}$ such that $\Gamma \vdash \varphi$ and $\mathcal{M} \models \Gamma^+$ but $\mathcal{M} \not\models \varphi^+$.) Note that if your answer is **yes**, you can work either with A(RC) or $\text{A(RC)}'$, as both give equivalent notions of derivability $\vdash$. But you should be clear which one you're choosing.